Unit 9: Multimedia Metadata, Organisation and Retrieval

- Metadata describes and manages resources so users and systems can find, interpret, connect and maintain them.
- A resource may be an image, sound recording, video, webpage, scanned letter or complete digital collection.
- BLISS emphasises discovery and access; BRIM adds provenance, series context, authenticity and controlled access.

TAKEAWAY A file without context may be stored but is difficult to discover or trust.

Learning outcomes and evidence

- Explain descriptive, structural, technical, administrative and preservation metadata with examples.
- Apply core Dublin Core elements and select local fields relevant to libraries or archives.
- Design identifiers, collection relationships, controlled keywords and a basic quality-check procedure; test searches.

TAKEAWAY Assessment should use a realistic catalogue record, not definitions alone.

Why multimedia needs metadata

- An image filename does not tell a user who created the image, when it was made or whether public use is permitted.
- Audio needs recording context, participants or roles, transcript availability, language and rights.
- Video needs title, creator, duration, captions, subject, access conditions and technical details.

TAKEAWAY **Metadata turns an isolated file into a resource that can be managed and found.**

Data, information, content and metadata

- Content is the resource itself—for example, the pixels of a photograph or samples in a WAV file.
- Metadata is structured information about that content, its context, use, technical properties or lifecycle.
- Metadata can be embedded in a file, stored in a catalogue database, linked in a manifest or kept in a preservation log.

TAKEAWAY **Do not assume all metadata is embedded in the media file.**

Five practical categories of metadata

- Descriptive: title, creator, subject, description and dates for discovery.
- Structural: sequence, part–whole links, page order and relationships between master and derivative.
- Technical: format, dimensions, duration and codec; administrative: ownership and access; preservation: fixity and events.

TAKEAWAY **One field can support several purposes; these categories overlap.**

Descriptive metadata in practice

- A meaningful title identifies the resource: “Student orientation guide”, not “final_video2”.
- Creator identifies responsible person or organisation; subject terms express what the resource concerns.
- Language, date, description and relevant geographic coverage improve filtering and interpretation.

TAKEAWAY **Describe the resource faithfully, using evidence rather than assumptions.**

Structural metadata and context

- A digitised 12-page file needs a reliable page order; an audiobook connects chapters and audio segments.
- A photograph might have a capture master, an access JPEG and a thumbnail, all belonging to one described item.
- Archival records also belong to series and fonds, not merely folders of unrelated scans.

TAKEAWAY Preserve meaningful relationships between resources and their parts.

Technical metadata in practice

- Images: pixel dimensions, colour space, bit depth and encoding format where needed.
- Audio: sample rate, channels, duration, bit depth or codec; video: frame size, frame rate, container and codec.
- Record technical facts from reliable file inspection and capture logs; do not guess based on an extension.

TAKEAWAY **Technical metadata helps render, migrate and evaluate media.**

Administrative and rights metadata

- Rights information records ownership, licences, restrictions and permission status—not simply “copyrighted”.
- Administrative information can record custody, acquisition, responsibility, embargoes and access procedures.
- Confidential records require enforceable access controls; an “internal” tag alone does not secure a public web file.

TAKEAWAY Describe restrictions accurately and enforce them outside metadata.

Preservation metadata and events

- Preservation metadata supports integrity, provenance and long-term management.
- A checksum algorithm and value can be stored with a verified date; a later check becomes a recorded event.
- PREMIS describes objects, events, agents and rights so actions on digital objects can be interpreted.

TAKEAWAY A checksum match supports byte-level fixity, not the truth of descriptive content.

Metadata standards: why interoperability matters

- Standards let systems exchange or interpret fields using shared meanings.
- A local field “item_description” may map to Dublin Core description for basic exchange.
- An institution can adopt a standard plus local rules for scope, required fields, vocabulary and controlled access.

TAKEAWAY A standard is a foundation; a useful application profile makes it operational.

Dublin Core: fifteen core elements

- Contributor, Coverage, Creator, Date, Description, Format, Identifier, Language, Publisher.
- Relation, Rights, Source, Subject, Title and Type complete the original fifteen elements.
- These elements are general and repeatable where appropriate; local systems may add more specific refinements.

TAKEAWAY The fifteen elements provide a shared vocabulary, not a complete archival management system.

Dublin Core example: photograph

- Title: “Fictional campus library opening”; Creator: Learning Media Lab; Date: 2024-06-12.
- Type: StillImage; Format: image/tiff; Subject: Libraries—Orientation; Language: not applicable to a purely visual photo.
- Identifier: LIB-PH-004; Rights: classroom demonstration, approved internal use.

TAKEAWAY **Record only known facts, and keep identifiers consistent across files.**

Dublin Core example: archival audio

- Title: “Fictional service desk oral history”; Creator: Training Media Team; Date: 2025-02-11.
- Type: Sound; Format: audio/wav; Subject: Records services; Language: sw; Identifier: ARC-OH-002.
- Relation: ARC-OH-002-transcript; Rights: internal training only; Description: synthetic classroom narration.

TAKEAWAY **Media and transcripts should be connected with explicit relationships.**

Choose fit-for-purpose richer standards

- MARC 21 supports rich bibliographic cataloguing; EAD is commonly used for archival finding aids.
- METS can package descriptive, administrative and structural information for complex digital objects.
- PREMIS focuses on preservation actions and entities; none of these should be confused with the fifteen Dublin Core elements.

TAKEAWAY **Select standards according to the service and interoperability requirements.**

Identifiers: unique, stable and meaningful

- A local identifier such as LIB-IMG-014 can distinguish an item within an institution.
- Separate item identifiers from file names and versions; a format conversion should not silently create a new intellectual item.
- Persistent identifier services such as DOIs or ARKs have governance requirements; a local code is not automatically persistent.

TAKEAWAY An identifier is useful only when its uniqueness and resolution are managed.

Design a file naming policy

- Example: ARC032_p003_master.tif and ARC032_p003_access.jpg connect page, role and item.
- Use predictable characters, consistent zero padding, roles and controlled versioning.
- Do not place passwords, private names or sensitive descriptions in public filenames.

TAKEAWAY **File naming supports management but is not a replacement for a catalogue.**

Collection hierarchy and archival context

- Example BLISS: collection → exhibition theme → digital item → media file.
- Example BRIM: fonds/creating body → series → file → item → digital representation; actual structures vary by institution.
- Describe provenance, original order and custody so records are not removed from their evidential context.

TAKEAWAY A search result must lead back to its collection and source context.

Relationships and derivatives

- Relation fields connect a master with its access JPEG, thumbnail, caption or OCR text.
- Record source identifiers and derivative events so a transformed file remains traceable.
- A transcript adds searchable content but should not be mistaken for the entire sound recording.

TAKEAWAY **Link related files and indicate their different functions.**

Subjects, keywords and controlled vocabularies

- Free keywords are flexible, but “student orientation” and “induction” may retrieve different results.
- A controlled vocabulary supplies preferred terms, variants and consistent subject labels.
- Local Swahili/English subject mappings can improve multilingual discovery if reviewed for meaning.

TAKEAWAY Search quality depends on consistent terminology and user language.

Authority control and names

- The same creator may appear as “Campus Library”, “Main Campus Library” or an abbreviation.
- Maintain a preferred name form, stable authority ID and known alternatives.
- Do not merge different people or organisations solely because their names look similar.

TAKEAWAY Authority records reduce ambiguity without erasing distinctions.

Dates, languages and missing values

- Use an agreed date convention, often ISO 8601 (YYYY-MM-DD) when full dates are known.
- Use explicit language codes consistently where appropriate, such as en or sw, after confirming content language.
- Unknown is different from an empty field, guessed date or publication date substituted for event date.

TAKEAWAY Good metadata preserves uncertainty instead of inventing precision.

Search: exact matches, text matches and facets

- Exact identifier search retrieves a known record; keyword search matches text in selected fields.
- Facets let users narrow results by type, subject, date range, language or permitted access.
- A query against titles only differs from one against title, description and subject.

TAKEAWAY **Define what is indexed and how users can filter the results.**

Full-text OCR versus catalogue metadata

- OCR extracts text from page images and can add useful searchable content.
- OCR errors, reading order and low-quality captures can produce false or missed matches.
- Metadata describes the whole resource and can remain useful when OCR is absent or unreliable.

TAKEAWAY **Use OCR as an additional access layer, not a replacement for description.**

Precision, recall and relevance

- Precision asks: of the returned results, how many are relevant to the query?
- Recall asks: of all relevant items in the collection, how many were returned?
- A narrow subject search may improve precision but miss synonyms; broader searching can increase recall and noise.

TAKEAWAY **Test with realistic tasks, not only whether the search box returns something.**

Metadata quality dimensions

- Completeness: required fields present; accuracy: facts match sources; consistency: shared rules followed.
- Uniqueness: identifiers are not duplicated; timeliness: rights and availability are kept current.
- Provenance: contributors, source evidence and revisions can be traced.

TAKEAWAY A filled field is not necessarily a correct field.

Quality assurance: review a batch

- Validate duplicate item IDs, missing titles, unknown rights states and inconsistent format names.
- Review subject terms, impossible dates, broken relations and publication restrictions with human judgement.
- Track corrections in an audit log; preserve prior meanings when changes affect record evidence.

TAKEAWAY Automated validation flags problems; people make contextual decisions.

Record structure: CSV and JSON

- CSV is convenient for flat tabular metadata; quote commas correctly and use UTF-8 encoding.
- JSON can express arrays and nested relationships but still requires an agreed schema.
- The schema should say which fields are required, repeatable, controlled, private and displayable.

TAKEAWAY A file format is not the same thing as a metadata standard.

Worked example: a catalogue row

- ID LIB-IMG-014 | Title “Local history training poster” | Type StillImage | Subject local history.
- Creator Training Media Team | Date 2025-01-10 | Rights classroom use | Access public demo.
- Related files: LIB-IMG-014_master.tif and LIB-IMG-014_access.jpg; each file has its own technical metadata.

TAKEAWAY A complete row names the item and points to its actual files.

Worked example: restricted record

- ID ARC-MIN-007 | Title “Fictional committee minutes” | Creator Training Records Office.
- Series ARC-MIN | Date 2024-05-17 | Type Text | Rights restricted internal use; access not public.
- A public discovery catalogue may show a limited description while the underlying files remain authorised-only.

TAKEAWAY **Public discovery and permission to view the content are separate decisions.**

Practical workflow: from file to retrieval

- 1. Inventory; 2. assign stable IDs; 3. capture known fields; 4. identify rights and sensitivity.
- 5. establish controlled subject terms; 6. connect related files; 7. validate; 8. test search/facets.
- 9. correct and record changes; 10. publish only approved descriptions and access files.

TAKEAWAY A search interface is the visible end of a documented metadata workflow.

Design an application profile

- Choose required fields (identifier, title, type, rights/access) and optional fields (creator, date, language, duration).
- State accepted values, missing-value rules, name authorities, identifiers and relationship rules.
- Include public-display rules and privacy review before exporting metadata.

TAKEAWAY An application profile makes a general standard consistent for one institution.

Hands-on retrieval task

- Search the fictional eight-record collection by keyword, type and access status.
- Compare a subject search against a broad title/description/subject query; record the number of results.
- Fix one incorrectly described record and explain how the correction improves retrieval or reduces risk.

TAKEAWAY **Measure the effect of metadata choices using realistic questions.**

BLISS application: library digital collection

- Describe a themed set of images, recordings and learning videos with stable titles and subjects.
- Connect original media to captions, transcripts and access versions.
- Use catalogue facets to help new users find a resource by topic, format, language and audience.

TAKEAWAY Organised metadata supports discovery and inclusive information services.

BRIM application: records and archives

- Preserve creating body, provenance, record series, original order and record status.
- Record relationships between files, page sequences, derivatives and preservation events.
- Distinguish discoverable descriptions from protected records; document who approves access.

TAKEAWAY **Trustworthy retrieval depends on archival context and appropriate controls.**

Summary and assessment checklist

- You should be able to classify metadata, populate a realistic Dublin Core-based record and justify local extra fields.
- Check unique identifiers, rights, dates, subject terms and linked derivatives using a simple validator.
- Test a relevant library/archive query and explain one improvement to precision, recall, usability or privacy.

TAKEAWAY The goal is dependable discovery with preserved context and responsible access.